



Ozgur Turkoglu

Middle East Technical University — Department of Statistics

ozgurtrko@gmail.com

+90 (543) 299 77 69

SUMMARY

Statistics senior at Middle East Technical University with hands-on experience in Python, R, SQL, Excel, and Tableau. Skilled in statistical modeling, machine learning, data analysis, and data visualization. Passionate about uncovering insights from complex datasets, especially in sports and business analytics. Strong teamwork and leadership skills developed through academic projects and experience as a basketball team captain.

EDUCATION

Middle East Technical University, Ankara

2019 - Present

Bachelor of Science in Statistics

- Solid foundation in statistical theory, data analysis, and applied modeling with emphasis on machine learning and data-driven decision-making.
- Completed group projects focused on real-world data interpretation, regression modeling, and statistical computing.
- Applied R and Python in academic and personal projects, including churn prediction and sports analytics.

EXPERIENCE

ConsciESG

July 2025 – October 2025

Data Analyst Intern

- Collected, cleaned, and structured ESG (Environmental, Social, and Governance) datasets; ensured data accuracy through validation checks and quality control for sustainability reporting.
- Conducted advanced data analysis and exploratory modeling using Python, R, and Excel to evaluate corporate ESG performance and identify improvement opportunities.
- Designed visual dashboards and concise summary reports, effectively communicating insights and recommendations to management to support data-driven ESG strategies.

PROJECTS

Kasko Conversion Prediction Model | Datathon Project

- Developed an end-to-end machine learning pipeline to predict insurance policy conversion using a dataset of 44,000+ records with severe class imbalance (~8.6% positive rate)
- Designed a leakage-free pipeline with time-series aware and group-based cross-validation to ensure realistic model evaluation
- Built a stacking ensemble model combining LightGBM and Gradient Boosting, achieving ROC-AUC: 0.77 and Average Precision: 0.315
- Applied Bayesian hyperparameter optimization (Optuna) to improve model performance and generalization
- Implemented SHAP-based feature selection and interpretability analysis to identify key drivers of conversion
- Conducted Lift and Gain analysis, enabling up to 4x improvement in customer targeting efficiency for sales teams
- Engineered domain-specific features (e.g., price ratios, interaction terms) to enhance predictive performance

Sampling and Survey Techniques Term Project

- Designed and conducted a survey of 278 METU students to analyze daily habits and campus sports center usage, ensuring representative sampling and reliable data collection.
- Applied statistical methods for **data cleaning, exploratory analysis, and hypothesis testing** to identify key factors influencing students' physical activity levels.
- Visualized results through dashboards and analytical summaries, translating findings into **actionable insights** on facility utilization and barriers to an active lifestyle.

Music & Mental Health – Multivariate Statistical Analysis Project (R)

- Analyzed a real-world survey dataset on music habits and mental health using advanced multivariate methods.
- Applied **MANOVA** to test group differences across multiple psychological outcomes.
- Used **PCA** and **Principal Components Regression** to reduce dimensionality and model relationships among 28+ predictors.
- Performed **Factor Analysis** to identify latent behavioral and mental-health structures.
- Built **LDA** models for group discrimination and visualized separability in discriminant space.
- Implemented **K-Means clustering** to uncover behavioral profiles in high-dimensional data.
- Conducted **Canonical Correlation Analysis** to quantify relationships between music-related and mental health variables.

University Score Prediction – Statistical Modeling & Data Mining Project (Python)

- Built a complete data mining pipeline from raw data to validated predictive models.
- Performed data cleaning, preprocessing, missing value handling, and exploratory analysis on real-world datasets.
- Engineered features and applied **dimension reduction (PCA)** to manage high-dimensional data.
- Trained and compared multiple models including **Linear Models, Regularized Regression, and Tree-based methods**.
- Used **cross-validation** and performance metrics (Accuracy, RMSE, R^2 , etc.) to evaluate and select models.
- Interpreted model outputs and translated technical results into clear, data-driven insights.

TECHNICAL SKILLS

Programming & Tools: Python, R, SQL, MS Excel, Tableau, R Shiny, Minitab

Statistical & Analytical Techniques: Multivariate Analysis (MANOVA, PCA, Factor Analysis, CCA), Regression & Regularization, Hypothesis Testing, Survey Design & Analysis, Dimensionality Reduction, Classification & Clustering, Linear Models, Stochastic Processes, Markov Chains, Market Basket Analysis

Data Science & Machine Learning: Data Cleaning and Manipulation, Feature Engineering, Predictive Modeling, Churn Analysis, Sports Analytics, Web Scraping, Data Visualization

CERTIFICATES

- Kaggle – Intro to Machine Learning
- Kaggle – Pandas
- Kaggle – Data Visualization
- Kaggle – Data Cleaning

LANGUAGES

English - Advanced (C1)

Turkish - Native

MEMBERSHIPS

Member, METU Statistics Club